

BEIHAI, China

WMO: 596440

Lat: 21.4566N

Lon: 109.1363E

Elev: 52

StdP: 14.67

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	43.5	45.9	29.8	23.9	55.3	33.6	28.3	52.6	22.6	49.3	20.9	49.8	11.8	0	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	91.6	80.4	90.6	80.4	89.7	80.4	83.5	87.4	82.7	87.4	82.0	87.0	7.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.5	169.6	85.8	81.5	164.0	85.6	80.7	159.9	85.6	47.6	87.3	46.8	87.7	46.1	87.3	86.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.9	17.2	15.2	41.6	94.5	2.9	1.6	39.5	95.6	37.7	96.5	36.1	97.4	34.0	98.5		
(5)			38.0	84.0	2.5	1.1	36.1	84.7	34.6	85.4	33.2	86.0	31.4	86.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	74.4	59.1	62.1	67.5	75.6	81.2	84.2	84.5	83.8	82.1	78.1	70.9	62.6	(6)
(7)		DBStd	10.62	7.83	8.75	7.46	5.98	3.87	2.72	2.62	2.58	3.17	4.12	5.97	6.47	(7)
(8)		HDD50	26	15	8	0	0	0	0	0	0	0	0	0	3	(8)
(9)		HDD65	577	215	147	65	8	0	0	0	0	0	0	19	123	(9)
(10)		CDD50	8920	296	348	543	768	967	1025	1068	1047	963	872	628	395	(10)
(11)		CDD65	3997	31	67	143	326	502	575	603	582	513	407	198	50	(11)
(12)		CDH74	40430	31	150	526	2361	5339	7217	7630	7074	5580	3380	1026	116	(12)
(13)	CDH80	14816	0	8	51	515	1874	3114	3328	2859	1993	931	141	2	(13)	
(14)	Wind	WSAvg	7.3	8.5	8.1	7.7	7.2	6.7	7.0	7.4	6.3	6.7	7.0	7.4	7.9	(14)
(15)	Precipitation	PrecAvg	67.6	1.5	1.6	2.3	3.9	5.6	10.3	14.3	14.6	7.5	3.3	1.3	0.7	(15)
(16)		PrecMax	87.1	8.4	8.2	7.4	7.8	28.3	23.0	36.3	31.6	14.1	8.1	8.6	3.8	(16)
(17)		PrecMin	33.5	0.0	0.0	0.3	0.4	0.9	2.6	2.4	4.6	0.9	0.4	0.0	0.0	(17)
(18)		PrecStd	11.7	1.9	1.6	1.9	2.3	5.0	5.7	8.4	6.6	3.3	2.1	1.8	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.1	81.3	83.8	88.4	91.5	92.5	92.6	93.4	91.9	89.9	85.5	79.8	(19)
MCWB			68.9	72.8	74.3	78.0	79.7	81.2	80.9	79.8	78.9	76.6	74.4	71.5	(20)	
2%		DB	74.5	77.9	81.5	86.6	89.8	90.7	91.1	91.3	90.3	88.0	83.5	77.1	(21)	
		MCWB	67.6	70.8	73.6	77.4	79.8	81.1	81.1	80.7	78.9	76.1	73.3	69.5	(22)	
5%		DB	72.4	75.6	79.2	85.0	88.5	89.6	90.1	89.9	89.1	86.4	81.6	74.9	(23)	
		MCWB	66.0	69.4	72.7	76.8	79.3	80.9	81.0	80.8	78.9	75.4	72.1	68.0	(24)	
10%		DB	70.1	73.4	76.9	83.2	87.2	88.6	89.0	88.7	87.9	85.0	79.7	72.5	(25)	
		MCWB	65.0	68.3	71.5	76.1	78.8	80.8	80.9	80.7	78.6	74.8	71.2	65.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.9	73.5	76.0	80.0	82.6	84.2	84.3	84.6	83.4	81.0	78.1	74.5	(27)
MCD B			74.4	79.2	80.7	85.6	87.6	87.2	87.6	88.0	86.3	85.0	81.3	77.7	(28)	
2%		WB	69.2	72.0	74.6	78.8	81.5	83.1	83.1	83.3	82.0	79.1	76.6	72.3	(29)	
		MCD B	72.6	76.4	79.1	84.1	86.7	87.2	87.6	87.5	86.4	83.8	80.1	75.5	(30)	
5%		WB	67.7	70.8	73.6	77.9	80.7	82.5	82.4	82.4	81.1	78.0	75.4	70.0	(31)	
		MCD B	71.2	74.5	77.9	82.9	86.0	87.3	87.7	87.1	85.9	83.1	79.5	73.2	(32)	
10%		WB	66.1	69.3	72.5	77.0	80.0	81.8	81.8	81.6	80.3	77.1	74.0	67.5	(33)	
		MCD B	69.2	72.7	76.4	81.7	85.2	86.8	87.1	86.5	85.3	82.3	78.1	71.1	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.6	9.9	9.6	9.2	9.2	7.5	7.6	8.6	9.8	11.3	12.0	11.7	(35)
MCDBR			12.8	11.6	11.4	10.6	9.8	8.1	8.7	10.3	11.2	12.5	13.1	13.4	(36)	
MCWBR			6.8	5.6	5.4	4.3	3.8	3.2	3.2	3.8	4.0	4.4	5.4	6.6	(37)	
5% WB		MCDBR	11.1	10.9	10.8	9.8	8.8	7.3	7.6	8.7	9.6	10.6	10.7	10.8	(38)	
		MCWBR	7.0	6.0	5.9	4.7	4.0	3.6	3.5	4.1	4.2	4.2	5.6	7.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.610	0.649	0.804	0.758	0.554	0.525	0.508	0.572	0.620	0.715	0.599	0.616	(40)
taud		1.732	1.685	1.448	1.566	2.061	2.164	2.203	2.000	1.846	1.613	1.812	1.745	(41)		
Ebn at Noon		209	211	187	200	243	248	253	238	224	193	211	201	(42)		
Edh at Noon		68	75	99	89	54	48	46	57	66	80	62	65	(43)		
(44)	All-Sky Solar Radiation	RadAvg		803	894	1000	1275	1545	1509	1572	1540	1495	1379	1125	903	(44)
(45)		RadStd		174	157	141	147	135	200	186	88	90	82	145	142	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+0.77	N/A	N/A	N/A	N/A	N/A	+133
(47)	Regional (5 neighbors)	N/A	N/A	N/A	+0.56	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page